

Biophysics – Part 2

Parcial examen

Tarragona, 15 Jan 2024

Name:

Question 1 (2 points). Define chaos and chaos theory. How does chaos differ from randomness? Explain the concept of sensitive dependence on initial conditions. Why is it a fundamental aspect of chaotic systems?

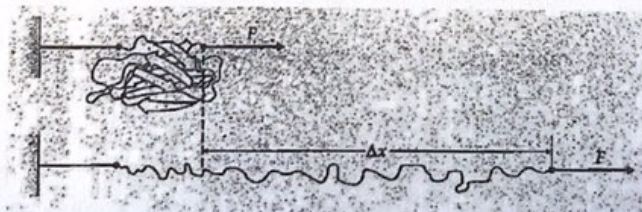
Question 2 (2 points). How do the molecular structures of liquid crystals contribute to formation of different liquid crystalline phases?

Question 3 (2 points). Explore the factors that influence the coil-globule transition in polymers. How does the polymer chain length, solvent quality, and interactions between monomers affect the transition?

Question 4 (2 points). Explain the concept of dynamic instability in microtubules. How does this property contribute to their role in cellular processes like mitosis?

Problem 1 (2 points). Unfolding a protein using atomic force microscope. Consider the polymer coil obeys Gaussian distribution of monomers,

$$P(r) \sim \exp\left(-\frac{3r^2}{2Na^2}\right)$$



Where r is the end-to-end distance and a is the size of a monomer. The interaction between monomers is controlled by second virial coefficient B such that $E = NkTBn$, where N is the length of the polymer and n is the density of monomers.

- Calculate the radius of the coil as a function of the polymer length in equilibrium.
- An atomic force microscope can be used to unfold a protein as shown on a Figure. Estimate the spring constant, assuming entropic nature of elasticity.